WESkit: Workflow Execution for HTC and Cloud

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GERMAN CANCER RESEARCH CENTER IN THE HELMHOLTZ ASSOCIATION

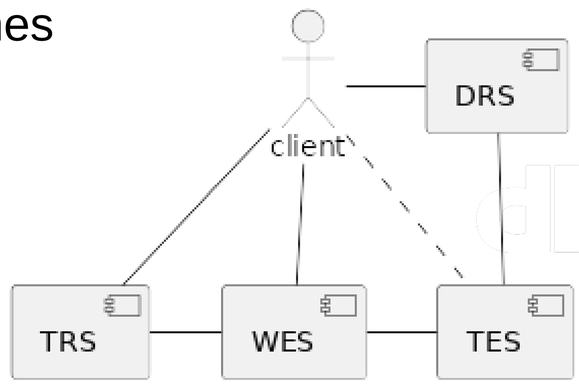
Research for a Life without Cancer

WESkit

- *REST service to execute workflows*
- *Compliant to the GA4GH WES specification*
- *Stable, robust, & scalable*
- *On-premise & in the cloud*
- *Open Source @ Gitlab.com*
- *Contributions from 4 institutes and in the context of ELIXIR::GA4GH*

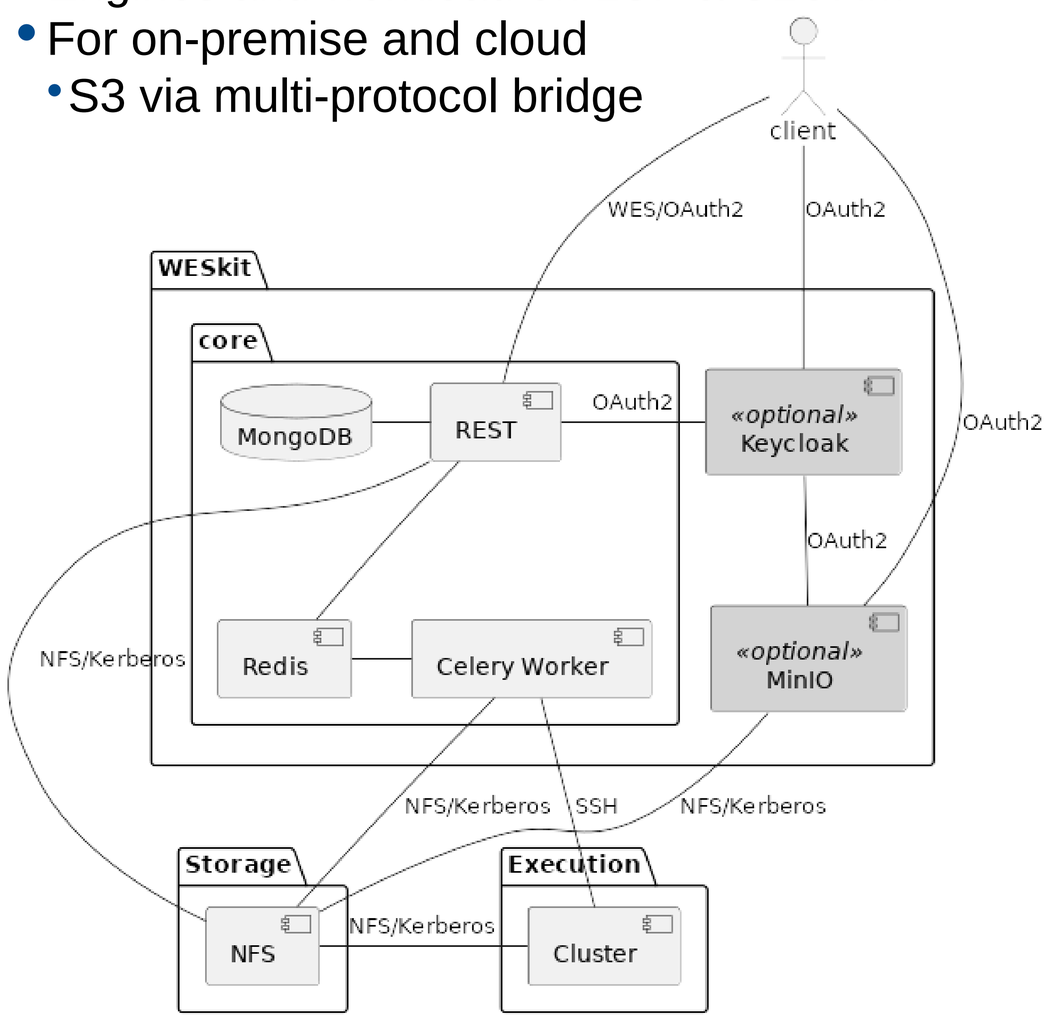
What is a Workflow Execution Service?

- Tool diversity = opportunities & problems!
  - Workflow engines (Nextflow, Snakemake.)
  - Computing (LSF, SLURM, Kubernetes)
  - Storage (NFS, S3)
- The GA4GH Workflow Execution Service (WES) API defines *uniform* access to different workflow engines



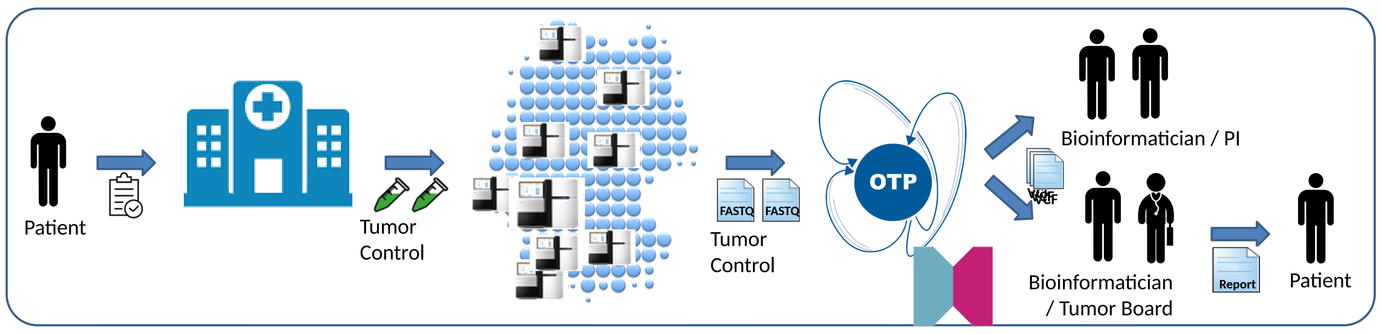
Requirements & Design

- Multiple workflow engines (NFL, SMK)
  - Shared filesystem
  - Native engines are used
- Failure tolerance and robustness
  - Stateless server & persistence & MongoDB
  - Distribution with Celery+Redis
  - All components can be replicated
- Engines and workload on LSF & SLURM
- For on-premise and cloud
  - S3 via multi-protocol bridge



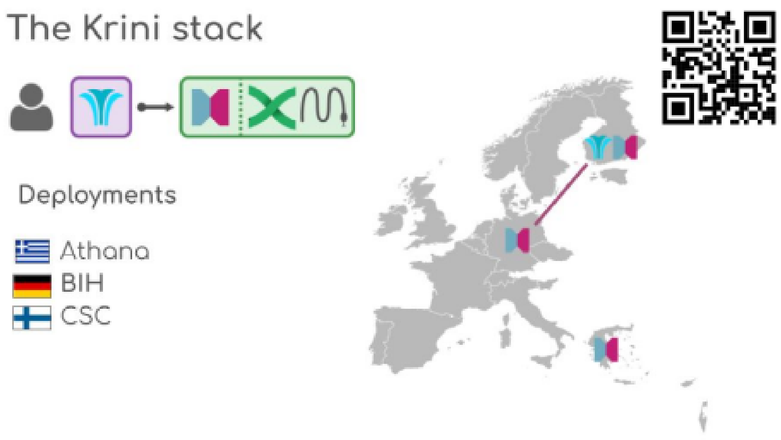
WESkit for Cancer Diagnosis @ DKFZ & BIH

Workflow execution component of the metadata management and workflow orchestrator One-Touch Pipeline (OTP)



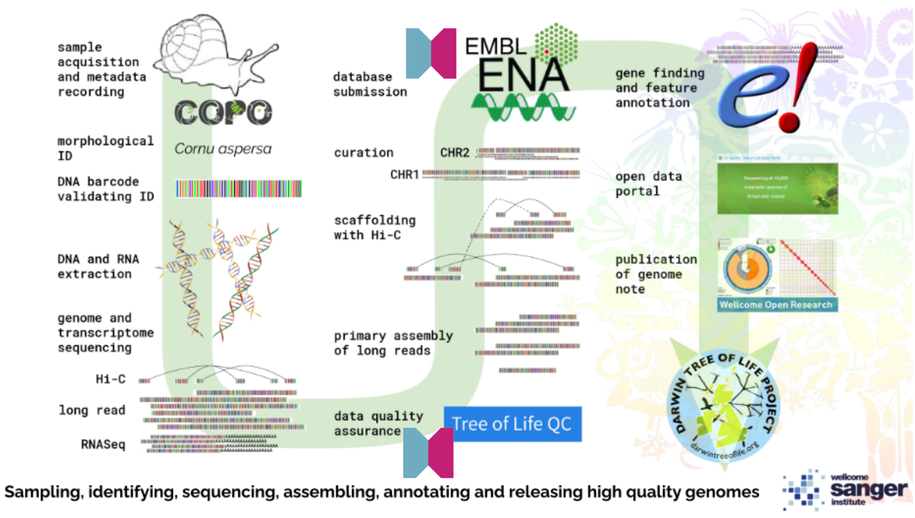
WESkit for Multi-Cloud Workflow Execution in Krini

Krini is a virtual research environment to run, manage, share, and publish life-science workflows.



WESkit for the Darwin Tree of Life Project

Multiple pipelines for WGS alignment, QC, phylogenetics, and data submission.



Next Milestones

- Kubernetes engine executor
- Improve failure robustness
- Production deployments at DKFZ and BIH
- CWL workflow engine
- Production-level TRS support
- Newer WES API specification versions

References

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Public demo @ https://weskit.bihealth.org/ga4gh/wes/v1/service-info
Public demo dashboard @ https://weskit.bihealth.org
Krini repository @ https://github.com/elixir-cloud-aai/krini
OTP repository @ https://gitlab.com/one-touch-pipeline/otp

The WESkit Git repositories can be found at https://gitlab.com/one-touch-pipeline/weskit

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